Mooketsi Vincent Magwaza

Full-Stack Engineer · Backend, Product Systems & Platform Engineering
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PROFESSIONAL SUMMARY

Product-minded full-stack engineer who turns real operational problems into working software across interface, API, data, security, deployment, and observability. Experience spans HarvestFlow's agricultural marketplace and BioAmps clean-energy systems, alongside offline-first geospatial and inventory products. Works across TypeScript, Rust, Python, PostgreSQL, Docker, CI/CD, and Grafana-backed operations.

EDUCATION

University of Botswana 2025 - Present

Computer Science | Gaborone, Botswana

EXPERIENCE

HarvestFlow - Full-Stack Engineer 2026 - Present

Agricultural marketplace | React 19, TypeScript, Tauri 2, Rust/Axum, PostgreSQL, Redis

- Developed an offline-aware Botswana marketplace connecting farmers, buyers, input suppliers, and logistics providers through produce listings, server-priced orders, transport offers, trip milestones, and append-only settlement records.
- Built React/Vite/Tauri clients and a Rust/Axum API with PostgreSQL and Redis; added role-based OTP onboarding, bounded offline caches, queued GPS check-ins, an admin operations console, Docker environments, Kubernetes manifests, and Prometheus/Grafana monitoring.

BioAmps Pty Ltd - Full-Stack Engineer 2025 - Present

Clean-energy software | React, TypeScript, Vite, Tailwind CSS, Supabase, IoT monitoring

- Developed the responsive BioAmps company platform covering its mission, clean-energy projects, gallery, contact, newsletter verification, and Energy Africa Conference content.
- Contributed to the Mokgosi biogas monitoring system: user and admin dashboards for sensor readings, trends, alerts, generator status, fault events, and automation controls across edge, aggregator, and cloud-facing workflows.

SELECTED PROJECT EXPERIENCE

Kori - Offline-First Wildlife Platform 2026

Product design, full-stack engineering, geospatial data, privacy, and operations

- Built four Next.js/TypeScript apps for low-connectivity safari use: a phone field app, landing site, review console, and API reference; sightings queue in IndexedDB and sync when connectivity returns.
- Implemented MapLibre/PMTiles offline maps, QR sharing, photo-metadata removal, delayed coarse location sharing, and track-only routing; a private Savuti satellite pack passed 16/16 offline checks.
- Developed a tested FastAPI/PostGIS backend with migrations, accounts, scoped API keys, uploads, reviews, trips, lifecycle jobs, Docker Compose, OpenTelemetry, Prometheus, Grafana, Tempo, backups, and CI.

Tagwise - Browser-Based Asset, Stock & Expiry Register 2026

Product strategy, React application, spreadsheet workflows, Rust services, and data integrity

- Started from a pharmacy problem: my sister needed a practical way to track batches, bulk stock, and expiry dates without buying a desktop POS subscription plus separate extensions.
- Built a responsive React/TypeScript workspace with an Excel-like register, QR/barcode scanning, labels, batch and expiry alerts, stock-takes, roles, history, conflict handling, and validated Excel/CSV import.
- Built five Axum services behind nginx with PostgreSQL, Redis, migrations, concurrency/idempotency tests, OpenAPI docs, Docker Compose, GitHub Actions, and OpenTelemetry/Prometheus/Grafana/Tempo.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Rust, Python, SQL, Go, Java
Frontend	React, Next.js, Vite, responsive UI, accessibility, IndexedDB
Backend	Axum, FastAPI, REST APIs, authentication, background jobs
Data	PostgreSQL, PostGIS, Redis, SQLAlchemy, sqlx, Alembic
DevOps & Observability	Docker, nginx, GitHub Actions, OpenTelemetry, Prometheus, Grafana, Tempo, Trivy
Native & Geospatial	Tauri 2, Windows APIs, WebView2, MapLibre, PMTiles